

Biila Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2022-2023
Mid-Semester Test (EC-2 Regular)

Course No. : AIMLCZG512

Course Title: Deep Reinforcement Learning

Nature of Exam : Closed Book Weightage : 30%

No. of Pages = 2 ;

No. Of Questions = 6;

Duration : 2 Hours;

Date of Exam: 21-07-2024 (EN)

Note to Students:

1. Answer all the questions.
2. Write your name and sign at the end of all the pages.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

We have considered your correct and alternative solutions even if it does not match the scheme given in this scheme

Q1) [Write the answer in an A4 Sheet. Scan & Upload]

- (a) How are rewards and returns connected? [1.5 M]

Answer key: Returns are the sum of (discounted) rewards.

$$G_t \doteq R_{t+1} + R_{t+2} + R_{t+3} + \dots + R_T$$
$$G_t \doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

Note: Details on what rewards are and will be ignored from the evaluation, except for the particular connection. One of the expressions is required or a clean statement connecting both is required. The correct answer gets 1.5. Penalty for vagueness.

- (b) Consider the RL agent learning the game of tic-tac-toe by playing against different randomly chosen opponents. Consider the temporal difference rule being used in this context. Can α be used to encourage exploration? [1 M] Why or why not? [2 M]

$$V(S_t) \leftarrow V(S_t) + \alpha [V(S_{t+1}) - V(S_t)]$$

Answer key: α cannot be used to encourage exploration. While using a constant α , the agent adapts to the changing environment, and when it is made to decrease with each step, the changes to the learned behavior become smaller with time. In either case, α cannot be used to encourage exploration. Hence, the answer is No.

Note: 1 Mark for a clear No as an answer with two marks for the explanation. Marks are deducted for incorrect reasoning/explanation.

- (c) Write one model-based and model-free algorithm for reinforcement learning covered in the course. [1.5 M]

Answer key: Algorithms covered under Dynamic Programming are model-based, including policy iteration, value iteration, etc. Algorithms covered under Monte Carlo are model-free as they do not assume the presence of model dynamics. **Note:** 1.5 marks for correctly answering both. deduct 0.5 for each mistake.

[1.5 + 3.0 + 1.5 = 6 Marks]

Q2) [Write the answer in A4 Sheet. Scan & Upload]

- (a) Consider again the RL agent learning the game of tic-tac-toe by playing against different randomly chosen opponents. It is decided that the learning algorithm does not use explicit exploratory moves? Under this scenario, would you consider RL agent to learn the same as as supervised learning? [One word, yes/no whose marking depends only on the correct explanation] [1 M] Explain in two well-articulated statements [Caution: your third and further statements for explanation will not be evaluated]. [2.0 M].

Answer key:

No.

Supervised Learning requires each move to be labeled which is not a requirement from reinforcement learning. In other words, supervised learning assumes moves are independent and fundamentally different from RL-based solutions.

Note: 1 Mark for No as an answer if the reasoning is correct. For the correct reasoning, 2 Marks.

- (b) State one situation in which a greedy action selection works better than ϵ -greedy action selection. [1.0 M].

Answer key:

If the rewards associated with actions are not stochastic, greedy action selection works better. This works once the rewards associated with the actions are uncovered.

Note: 1 mark for this case or an equivalent case.

- (c) Write any one use-case (application) from your workplace which can be modeled as a multi-armed bandit for its solution. Your answer should have all the elements that identify the use case and can be modeled as a multi-armed bandit problem. [vague description of application will be penalized]. [2 M]

Answer key:

A use case with no states, with n number of actions, and reward distribution aligns with MAB is required.

Note: Marks are awarded only if the case is appropriate to be modelled as MAB, and the answer brings it out.

[3 + 1.0 + 2 = 6 Marks]

Q3) [Write the answer in A4 Sheet. Scan & Upload]

- (a) How do you use the following learning rule for a non-stationary MAB problem? Explain in two well-articulated statements [Caution: your third and further statements for explanation will not be evaluated]. [2.0 M].

$$Q_{k+1} = Q_k + \alpha [R_k - Q_k]$$

Answer key:

- For a non-stationary MAB problem, the learning rule should account for the changing nature of the environment by **giving more weight to recent rewards**.
- Fixing the learning rate constantly automatically emphasizes adapting the recent behavior.

Note: It is very important to note that constant alpha implies more weightage to recent behavior. 2 marks for the correct answer.

- (b) Consider encouraging exploration by Optimistic Initial Values and UCB. Provide one scenario when the approach is useful and one when they are not, in a neat table as below:

	When it works?	When it doesnt work?
Optimistic Initial Values	[0.5 M]	[0.5 M]
UCB	[0.5 M]	[0.5 M]

Answer key:

	When it works?	When it doesnt work?
Optimistic Initial Values	When initial exploration is crucial, and the environment has sufficient variability to guide the agent towards optimal actions through exploration . [0.5 M]	When the initial optimism is misleading and there is not enough exploration (exploitation) to correct over-optimistic initial estimates, leading to suboptimal long-term behavior. [0.5 M]
UCB	The exploration-exploitation trade-off needs to be managed efficiently in an environment where the variance of rewards can be estimated and utilized to guide exploration. [0.5 M]	In highly non-stationary environments where the assumptions of stable variance and confidence intervals are invalid , leading to inaccurate upper confidence bounds and poor decision-making. [0.5 M]

(c) What information do you need about an RL problem to find out the optimal value function? [2 M].

[2.0 + 2.0 + 2.0 = 6 Marks]

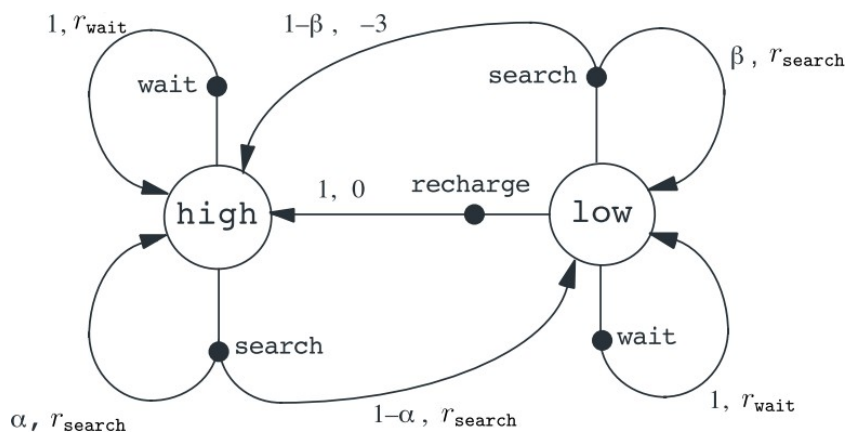
Answer key:

Information needed for RL problem to find optimal value function: State transition probabilities, reward function, discount factor (γ), policy (π):

Note: It is very important to note that constant alpha implies more weightage to recent behavior. 2 marks for the correct answer.

Q4) [Write the answer in A4 Sheet. Scan & Upload]

(a) Recollect the recycling robot example whose transition diagram is as below:



Write down the expression for $q_{\pi}(\text{high}, \text{search})$, and $q(\text{high}, \text{search})$. [3.0 M]. Assume the notations introduced in the class for interpreting the transition diagram.

Answer key:

(a)

$$q_{\pi}(high, search) = \sum_{\substack{s' \in \{high, low\}, \\ r \in \{r_{search}\}}} p(s', r | s, a) [r + \gamma v_{\pi}(s')]$$

① Expand this expression for all s', r .

$$q_{\pi}(high, search) = \sum_{\substack{s' \in \{high, low\}, \\ r \in \{r_{search}\}}} p(s', r | s, a) [r + \gamma \max_{a'} q_{\pi}(s', a')]$$

② Expand this expression for all s', r .

constant expansion, the expression in the form given here gets max 2 Marks.

- (b) Imagine that you are designing a robot to run a maze. You decide to give it a reward of +1 for escaping from the maze and a reward of zero at all other times. The task seems to break down naturally into episodes—the successive runs through the maze—so you decide to treat it as an episodic task, where the goal is to maximize expected total reward. After running the learning agent for a while, you find that it is showing no improvement in escaping from the maze. What is going wrong? How to fix the situation?

Answer key:

The agent is not penalized for staying longer without escaping the maze. It escapes in a few minutes, vs a few hours vs a few days; the total reward it gets is 1. It is possible that the agent does nothing better for a longer time. It is essential to penalize the agent for spending longer in the maze. If for every step the agent is in the maze gets a negative score, the only way to ensure a better score is to find a way out.

Note: Stating the reason correctly - 1.5 marks. Explaining how to fix the situation is 1.5 marks.

[3.0 + 3.0 = 6 Marks]

Q5) [Write the answer in A4 Sheet. Scan & Upload]

Consider the following episodes as experienced by an agent following a policy π with states {A,B,C,T} in which T is terminal state and actions {x,y,z}. The episodes are written using the representation { state - action - reward- nextState ... } order, and the number in red within the bracket is the immediate reward.

E1: A - x - (+10) - C - y - (+12) - C - x - (0) - T

E2: A - x - (+10) - C - x - (+2) - B - y - (+12) - C - x - (+ 20) - T

E3: C - x - (+2) - B - y - (+12) - C - x - (+ 20) - T

Assuming the initial $Q\pi(s,a)$ are all 0's, compute the revised $Q\pi(s,a)$ using on-policy first-visit MC control algorithm for ϵ -soft policies for one iteration only [4 M]. Assume the $\gamma = 0.8$. Write down the revised policy. [2 M].

[6 Marks]

Answer key:

[**Verify the numerical once**]

Episode1:

A-x-(+10):

$$G_0 = 10 + 0.8(12 + (0.8)(0)) = 19.6$$

C-y-(+12):

$$G_1 = 12 + (0.8 \cdot 0) = 12$$

C-x-(0):

$$G_2 = 0$$

Episode 2:

A-x-(+10):

$$G_0 = 10 + (0.8)(2 + 0.8(12 + (0.8)20)) = 29.52$$

C-x-(+2):

$$G_1 = 2 + (0.8)(12 + (0.8)20) = 24.4$$

B-y-(+12):

$$G_2 = 12 + 0.8(20) = 28$$

C-x-(+20):

$$G_3 = 20$$

Episode 3:

C-x-(+2):

$$G_0 = 2 + (0.8)(12 + (0.8)(20)) = 24.4$$

B-y-(+12)

$$G_1 = 12 + 0.8(20) = 28$$

C-x-(+20)

$$G_2 = 20$$

Updating Q values:

1) $Q(A,x)$

$$E1: G_0 = 19.6$$

$$E2: G_0 = 29.52$$

$$\text{Average} = (19.6 + 29.52) / 2 = 24.56$$

2) $Q(C,y)$

$$E1:G1=12$$

$$\text{Average}=12$$

3) $Q(C,x)$

$$E1:G2=0$$

$$E2: G1=24.4$$

$$E3:G0=24.4$$

$$\begin{aligned}\text{Average} &= (0+24.4+24.4)/3 \\ &= 16.93\end{aligned}$$

4) $Q(B,y)$

$$E2:G2=28$$

$$E3: G1=28$$

$$\text{Average}=28$$

Revised Policy is

For A, $Q(A,x)=24.56$

For C, $Q(C,x)=16.93$

$$Q(C,y)=12$$

We have to chose x, as it has higher Q value

For B, $Q(B,y)=28$

Thus, the Revised policy is

$$pi(A)=x$$

$$Pi(C)=x$$

$$Pi(B)=\text{Terminal}$$